



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY
PAPER 4 Option Topics

9189/4

NOVEMBER 2014 SESSION

1 hour 15 minutes

Additional materials:

Answer paper

Graph paper

Data booklet

Mathematical tables and/or electronic calculator

TIME 1 hour 15 minutes

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper.

Answer a total of **four** questions. Do not answer more than **two** questions from any **one** Option.

You are advised not to attempt questions on Options for which you have not been prepared.

Write your answers on the separate answer paper provided.

Begin each answer on a fresh page.

If you use more than one sheet of paper, fasten the sheets together.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

Where relevant, the symbols for aluminium, chlorine and iodine are printed as: *Al*; *Cl*; *I*; respectively.

You are reminded of the need for good English and clear presentation in your answers.

This question paper consists of 9 printed pages and 3 blank pages.

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BIOCHEMISTRY

Not more than two questions to be answered from this Option.

- 1 (a) (i) Define the term *primary structure of a protein*.
- (ii) State the groups that form
1. hydrogen bonds in a secondary structure,
 2. hydrogen bonds in a tertiary structure of a protein.

[3]

- (b) Spider silk, a hydrophobic protein, secreted by spiders contains a high proportion of glycine and alanine.

glycine: $\text{H}_2\text{NCH}_2\text{CO}_2\text{H}$
 alanine: $\text{H}_2\text{NCH}(\text{CH}_3)\text{CO}_2\text{H}$

- (i) Define the term *hydrophobic*.
- (ii) Explain why
1. spider silk is hydrophobic,
 2. glycine-rich regions of proteins give rise to their flexibility.

[4]

- (c) Spider silk has properties similar to nylon.

- (i) Explain why spider silk and nylon have similar properties.
- (ii) If spider silk could be manufactured industrially, suggest its possible use.

[2]

- (d) Suggest why there are seven different types of spider silks.

[1]

[Total: 10]

- 2 (a) State the processes that are involved in protein synthesis, giving the site where each process takes place. [4]
- (b) Haemoglobin is made up of two α -helical chains and two β -pleated chains.
- (i) Define a *gene*.
- (ii) State the number of genes that are required for the formation of haemoglobin.
- (iii) Describe the relationship between *mRNA* and a gene.
- (iv) Name the co-factor in haemoglobin and state its function.
- (v) Explain how the presence of carbon monoxide affects the function of haemoglobin.

[6]

[Total: 10]

- 3 (a) Name two membrane proteins and state the function of each. [4]
- (b) With the aid of full structural formulae, differentiate a triglyceride from a phosphoglyceride. [3]
- (c) Draw diagrams showing the arrangement of phospholipids when placed in
- (i) pure water,
- (ii) pure benzene,
- (iii) a mixture of water and benzene.

[3]

[Total: 10]

ENVIRONMENTAL CHEMISTRY

Not more than two questions to be answered from this Option.

- 4 (a) One method used to remove cadmium ions, Cd^{2+} , from waste discharged from metal plating waste is ion exchange.
- (i) Outline the steps involved in an ion exchange method.
- (ii) State the features of a resin that makes it suitable for use in ion exchange.
- (b) (i) Using equations, explain how aluminium ions enhance soil acidity.
- (ii) Explain why increased acidity causes lead ions to become more mobile.

[5]

[5]

[Total: 10]

- 5 (a) Fig. 1 shows stages followed in the treatment of water.

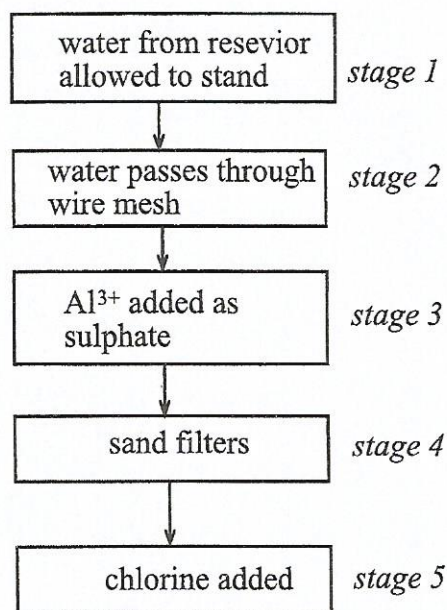


Fig. 1

- (i) Explain why the water should be allowed to stand in *stage 1*?
- (ii) Give the role of the aluminium ions in *stage 3*.
- (iii) State the function of chlorine in *stage 5* and write a balanced equation for the reaction of chlorine with water.

[6]

- (b) Dissolved carbon dioxide is used by algae for photosynthesis.
- (i) Write equilibrium reaction equation for carbon dioxide dissolved in water.
- (ii) Use the equilibrium reaction equations in (i) to explain how photosynthesis reduces water acidity.

[4]

[Total: 10]

6 (a) Define *Eutrophication*. [2]

(b) Describe with the aid of an equation, the method used to remove

- (i) phosphate ions from contaminated water.
- (ii) ammonium ions from contaminated sludge.

[4]

(c) Rocks are formed in oceans using dissolved carbon dioxide.

(i) Use equations to show

1. how the rocks are formed,
2. the buffering action of carbon dioxide.

[4]

[Total: 10]

PHASE EQUILIBRIA

Not more than two questions to be answered from this Option.

- 7 The distribution constant for solution X between *n*-hexane and H₂O is 9.6.
- (a) Calculate the final concentration of X when 25 ml of 0.05 mol dm⁻³ X is extracted with
- a single 25 ml portion of *n*-hexane,
 - two 12.5 ml portions of *n*-hexane.
- [6]
- (b) Identify the extraction in which a larger amount of X was extracted. [1]
- (c) (i) State and explain the effect of adding 2 g of NaCl_(s) to the 25 ml of 0.05 mol dm⁻³ X before the extractions.
- (ii) State any other factor that affects the value of the distribution constant.
- [3]
- [Total: 10]

- 8 (a) A mixture of alanine, aspartic acid and serine was analysed by thin layer chromatography (T.L.C), with SiO₂ as a stationary phase. The T.L.C mobile phase was 95% ethanol and 5% H₂O.

alanine: $\text{H}_2\text{NCH}(\text{CH}_3)\text{CO}_2\text{H}$

aspartic acid: $\text{H}_2\text{NCH}(\text{CH}_2\text{CO}_2\text{H})\text{CO}_2\text{H}$

serine: $\text{H}_2\text{NCH}(\text{CH}_2\text{OH})\text{CO}_2\text{H}$

- (i) Deduce with reasons the amino acid with the
- highest R_f value,
 - lowest R_f value.
- (ii) The stationary phase, adsorbed SiO₂, was mixed with silver ions, Ag⁺. State and explain how the magnitudes of the R_f values were affected by the Ag⁺ ions.
- [6]
- (b) Draw a labelled T.L.C chromatogram that can be used to show that all the three amino acids were present in the analysed mixture. [3]
- (c) State **one** other application of T.L.C. [1]

[Total: 10]

9 At one atmosphere, pure water boils at 100°C.

- (a) (i) State and explain how the boiling point of H_2O is affected by adding CCl_4 .
- (ii) Suggest with a reason how the boiling point of the $\text{H}_2\text{O}/\text{CCl}_4$ mixture is affected by lowering the pressure in the environment.

[6]

- (b) (i) State the best method for extracting *sea water solution*?
- (ii) Explain why steam distillation is used to extract a thermally unstable perfume from crushed flower petals.

[4]

[Total: 10]

TRANSITION ELEMENTS

Not more than two questions to be answered from this Option.

- 10 (a) (i) State **three** ways by which the addition of chromium to mild steel improves its properties.
- (ii) Describe the bonding in $[\text{Cr}(\text{H}_2\text{O})_6]^{3+}$ and state the bond angle. [5]
- (b) Describe and explain the colour changes that occur when orange $\text{Na}_2\text{Cr}_2\text{O}_7$ is added to
- (i) $\text{NaOH}_{(\text{aq})}$,
- (ii) $\text{Sn}^{2+}/\text{H}^+_{(\text{aq})}$. [5]
- [Total: 10]
- 11 (a) The oxidation of manganese (IV) oxide in molten potassium hydroxide with oxygen produces potassium manganate (VI), which when added to dilute acid disproportionate to a purple solution and a black solid, manganese dioxide.
- (i) State the colour of potassium manganate (VI).
- (ii) Identify the manganese containing ion in the purple solution.
- (iii) Write an equation to show the disproportionation of potassium manganate (VI).
- (iv) Suggest an equation for the production of potassium manganate (VI). [4]
- (b) State any **two** uses of manganese (IV) oxide. [2]
- (c) Describe and explain what would be observed when aqueous sodium hydroxide is added to an aqueous solution of manganese (II) until in excess and final mixture is exposed to air for sometime. [4]
- [Total: 10]

- 12 (a) Table 1 shows complexes of cobalt and some of their properties. Different proportions of chloride are precipitated when each of the complexes is treated with aqueous silver nitrate.

Table 1

Formula of complex	Colour of solid	Number of Cl ⁻ precipitated	Dipole moment
$\text{CoCl}_3(\text{NH}_3)_6$	orange	3	absent
$\text{CoCl}_3(\text{NH}_3)_5$	violet	2	present
$\text{CoCl}_3(\text{NH}_3)_4$	violet	1	present
$\text{CoCl}_3(\text{NH}_3)_4$	green	1	absent

- State the isomerism in the compounds.
- Suggest a structure for each complex.
- Explain the presence or absence of a dipole moment in each complex.

[8]

- (b) Cobalt, in the form of vitamin B₁₂, is needed for healthy functioning of the body.

Name the

- disease caused by the deficiency of vitamin B₁₂,
- cells in the body that are affected by the deficiency of vitamin B₁₂.

[2]

[Total: 10]